Futures / Forward : Obligations.

Long F : Long asset.

Short F : Short asset.

Long F.O.



Options : Right.

Long Call : Long asset.

Short Call : wait . . .

Long put : Short asset.

put put : wait . . .

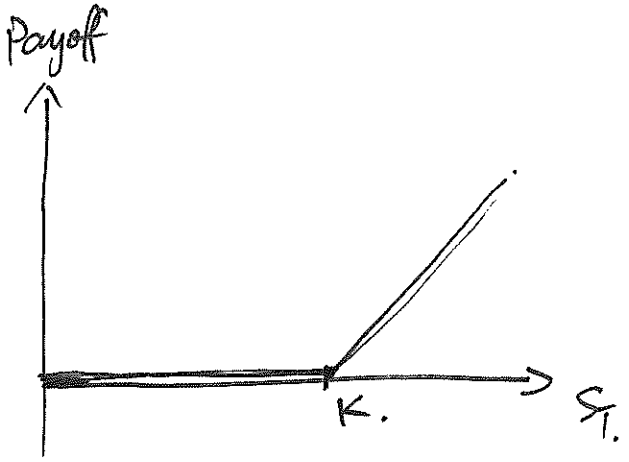
premium : C & P Strike price : K.

Long Call: right to buy asset.

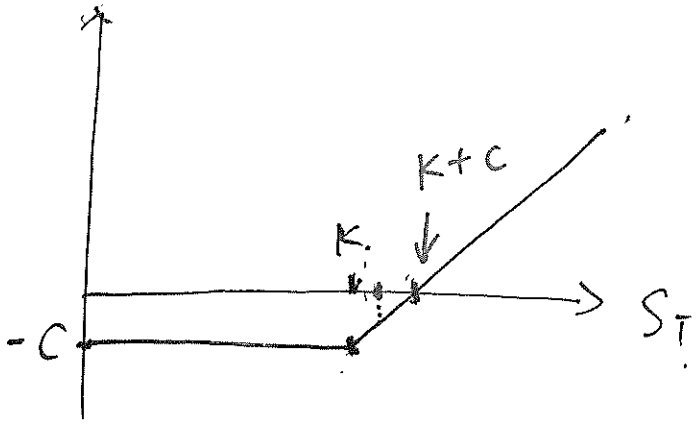
Payoff = $\max(S_T - K, 0)$.

if $S_T > K$ → exercise → Payoff = $S_T - K$.

if $S_T \leq K$ → no exercise → Payoff =



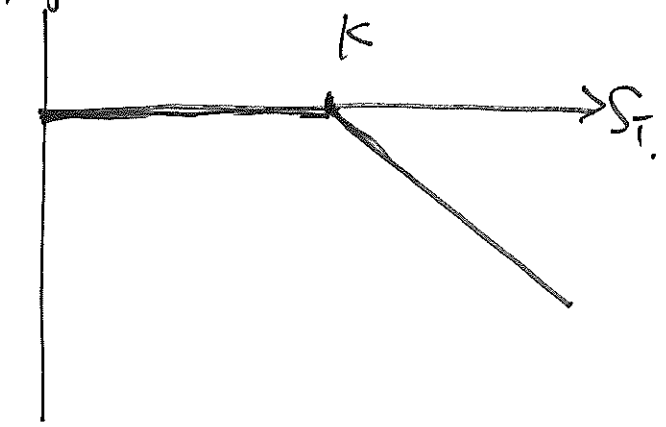
Profit = $\max(S_T - K, 0) - C$.



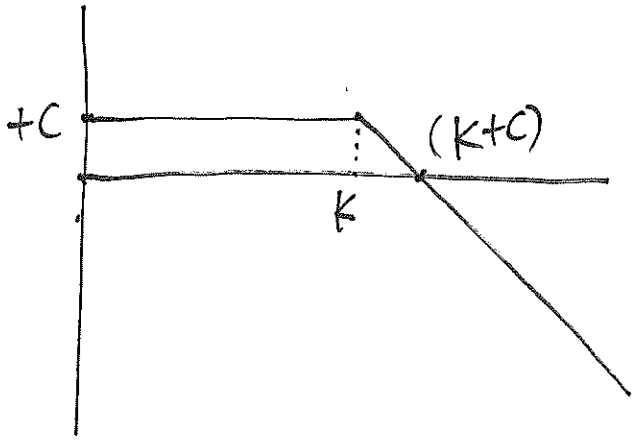
Short Call

Payoff = $\min(K - S_T, 0)$

Payoff.



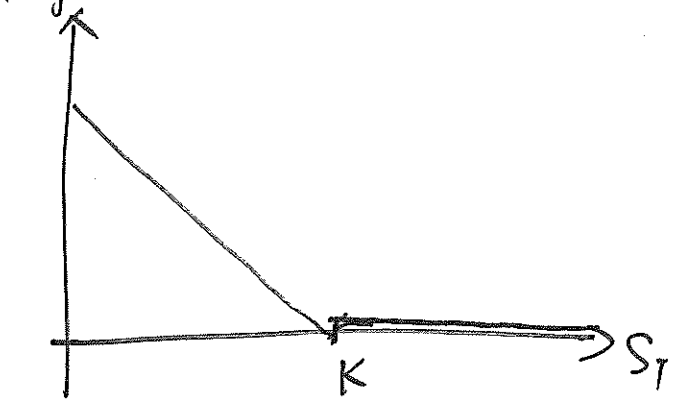
Profit = $\min(K - S_T, 0) + C$.



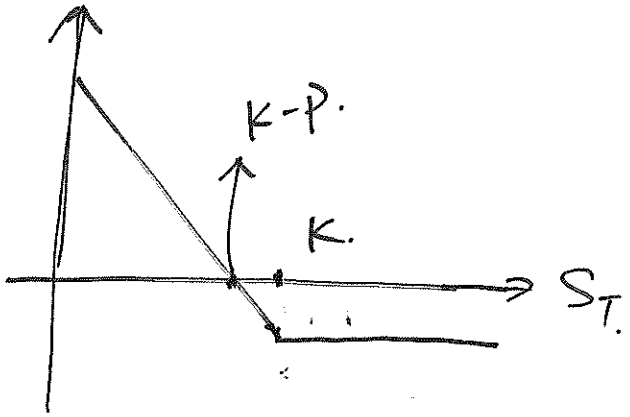
Long Put : right to sell asset.

Payoff = $\max(K - S_T, 0)$ { if $K > S_T$ → exercise → $K - S_T$
if $K \leq S_T$ → no exercise → 0

Payoff.



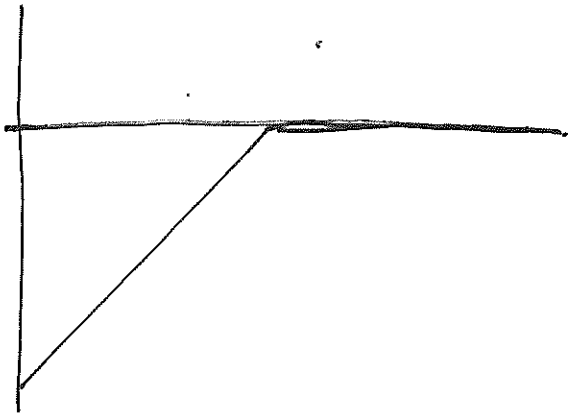
Profit = $\max(K - S_T, 0) - P$.

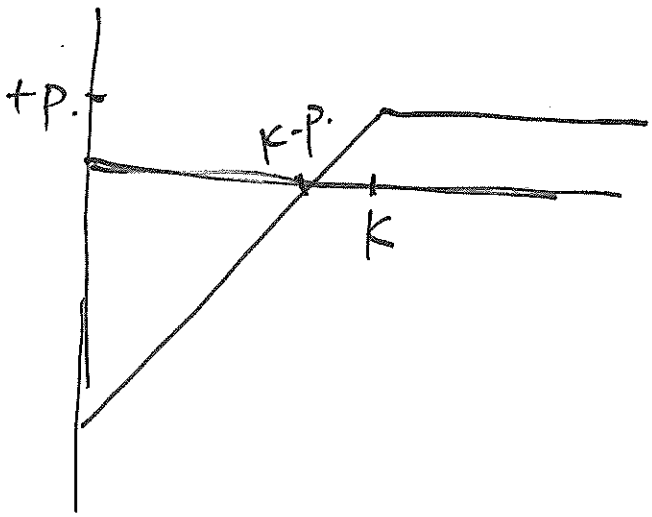


Short Put :

Payoff = $\min(S_T - K, 0)$.

Profit = $\min(S_T - K, 0) + P$.



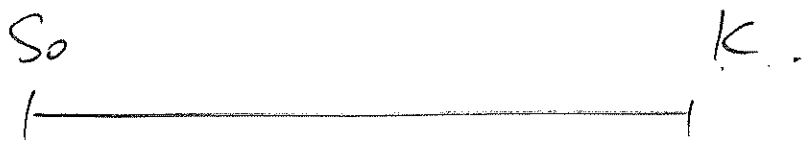


$C = \$25$ $S_0 = \$20$ $S_T = \$100$

$K = \$0$

Buy C. Profit = $100 - 0 - 25 = \$75$.

Buy $S_0 = \$20$ Profit = $100 - 20 = \$80$.



Lower bound for c on non-div paying stock.

$$c \geq \max(S_0 - K \cdot e^{-rT}, 0).$$

	0	T	
		$S_T > K$	$S_T \leq K$
A	c $K \cdot e^{-rT}$	S_T K	0 K
B	$c + K \cdot e^{-rT}$	S_T	K
B	S_0	S_T	S_T

At T $\begin{cases} S_T > K, & A = B. \\ S_T \leq K, & A \geq B. \end{cases}$ $\Rightarrow A \geq B.$

At t_0, $c + K \cdot e^{-rT} \geq S_0$

$$c \geq S_0 - K \cdot e^{-rT}$$

Lower bound for put on non-dividend ...

$$P \geq \max\left(K \cdot e^{-rT} - S_0,\ 0\right)$$

	to	T	
		$S_T \geq K$	$S_T < K$
C	P	0	$K - S_T$
	S_0	S_T	S_T
Total	$P + S_0$	S_T	K
D	$K \cdot e^{-rT}$	K	K

At T,

if $S_T \geq K$, $C \geq D$.
if $S_T < K$, $C = D$
} $C \geq D$.

At to, $P + S_0 \geq K \cdot e^{-rT}$

$$P \geq K \cdot e^{-rT} - S_0$$

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